1 Introduction

Ledrappier's 1999 paper *Distribution des orbites des réseaux sur le plan réel* [Led99] proves an equidistribution theorem for averages of compactly suported real functions on $\mathrm{SL}(2,\mathbb{Z})$ orbits. The goals of this document are twofold *(i)* rewrite Ledrappier's paper in the language of Teichmüller theory, to better motivate and understand the result, and *(ii)* to explicate terse elements of Ledrappier's proof.

The following notation will be used throughout

- $\mathbb{T}^2$ will denote the (topological) torus
- $\mathbf{T_1}$ will denote the Teichmüller space of $\mathbb{T}^2$
- Γ will denote the mapping class group of $\mathbb{T}^2$
- $\mathcal{Q}^1$ will denote the space of unit quadratic differentials over $\mathbf{T_1}$
- $\mathscr{MF}$ will denote the space of measured foliations on $\mathbb{T}^2$.
- μ_{Th} will denote the Thurston measure on measured laminations, which may be considered as a measure on measured foliations under the equivalence of measured laminations and foliations.
- $\mathcal{Q}^1\mathcal{M} = \Gamma\backslash\mathcal{Q}^1$ will refer to the moduli space of unit quadratic differentials.
- μ_{MV} will denote the Masur-Veech measure on $\mathcal{Q}^1$, we may also consider this as a measure on the quotient by considering a fundamental domain.
- Fix once and for all curves α, β with homology classes giving a basis for $H^1(\mathbb{T}^2;\mathbb{Z})$ and $i(\alpha,\beta) = 1$. Moreover, we may fix once and for all a covering map $\mathbb{R}^2 \to \mathbb{T}^2$, with (α,β) corresponding to the ordered lattice generators $((1,0),(0,1))$.

The equidistribution theorem of Ledrappier states that if $X \in \mathbb{R}^2$ is a vector having irrational slope, then for $f \in C_c(\mathbb{R}^2)$

$$\frac{4}{||X||\,\mathrm{Vol}(\mathrm{SL}(2,\mathbb{Z})\setminus\mathrm{SL}(2,\mathbb{R}))}\int_{\mathbb{R}^2}\frac{f(Y)}{||Y||}dY = \lim_{T\to\infty}\frac{1}{T}\sum_{\substack{g\in\mathrm{SL}(2,\mathbb{Z})\\ ||g||_F\le T}} f(gX)$$

The paper applies two reductions, first showing that it is sufficient to prove the result for $f \in C_c(\mathbb{R}^2\setminus\{0\})$ by bounding

$$\#\left\{g\in\mathrm{SL}(2,\mathbb{Z})||g||_F \le T \text{ and } ||gX|| < \epsilon\right\}$$

Then noticing that the result is trivial for odd functions, thus reducing to proving it for even functions. We can instead view an even function f with these properties as $f \in C_c(\mathbb{RP}^1 \times \mathbb{R}_{>0})$, which we will see can be interpreted as the parameterization for the set of measured foliations on the torus with respective components by a projective foliation and corresponding horizontal stretch. By recontextualizing the use of hyperbolic geometry in Ledrappier's proof as Teichmüller geometry, the elements of the proof become much more natural. In the Teichmüller geometry setting $\mathrm{SL}(2,\mathbb{Z})$ becomes identified with Γ $X \in \mathscr{MF}$ be a foliation with non-periodic leaves, then for $f \in C_c(\mathscr{MF})$

$$\frac{8}{||X||\mu_{\mathrm{MV}}(\mathcal{Q}^1\mathcal{M})}\int_{\mathscr{MF}}\frac{f(Y)}{||Y||}d\mu_{\mathrm{Th}}(Y) = \lim_{T\to\infty}\frac{1}{T}\sum_{\substack{\gamma\in\Gamma\\ \sqrt{K_\gamma}\leq T}} f(\gamma X)$$

2 Teichmüller Theory for the Torus

Definition 1. *The Teichmüller space of the torus $\mathbf{T_1}$ will be defined as the set of pairs (S, φ) where S is a genus 1 Riemann surface, and $\varphi : \mathbb{T}^2 \to S$ is an orientation preserving diffeomorphism, two pairs $(S_1, \varphi_1), (S_2, \varphi_2)$ will be identified if there is a biholomorphism $I : S_1 \to S_2$, such that $I\varphi_1$ is homotopic to φ_2.*

In order to work with Teichmuller space, I will identify it with the space of abelian differentials of the form $dx + \tau dy$ for $\tau \in \mathbb{H}$, I will do this by first identifying $\mathbf{T_1}$ with representatives of the form $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, then pulling back the flat structure on the torus, by using the fixed covering on $\mathbb{T}$.

Start with a point $(S, \varphi) \in \mathbf{T_1}$, then we get an ordered pair of $\pi_1(S)$ generators from our marking $\varphi_*(\alpha)$ and $\varphi_*(\beta)$. Considering the covering map $\mathbb{C} \to S$, we can consider the images of deck transformations $A = 0 \cdot \varphi_*(\alpha), B = 0 \cdot \varphi_*(\beta)$ so that S is isometric to $\mathbb{C}/(A\mathbb{Z} + B\mathbb{Z})$. Such an identification with lattices is only defined up to linear isometry in $\mathbb{C}$, thus only up to rotation and dilation.

Conversely, we may start with a flat torus $S = \mathbb{C}/(A\mathbb{Z} + B\mathbb{Z})$, with $\pi_1(S)$ generated by α', β' with deck transformations giving A, B as above. There is a diffeomorphism $\varphi : \mathbb{T}^2 \to S$ with $\varphi_* : \alpha, \beta \mapsto \alpha', \beta'$, thus identifying $\mathbb{C}/(A\mathbb{Z}+B\mathbb{Z})$ with a point in Teichmüller space. Isometries of lattices homotopic to the identity must be linear transformations acting trivially on $H_1(S, \mathbb{Z})$, thus two marked lattices are identified with the same point in $\mathbf{T_1}$ exactly when they differ by a rotation and dilation.

We can choose a unique representative of each set of lattices up to rotation and dilation by multiplying by complex number A^{-1}, thus identifying our lattices with $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$, for $\mathbb{Z} = A^{-1}B$. Since $\pi_1(S)$ acts by translation it is orientation preserving, thus the orientation on the universal cover $\mathbb{C}$ induces an orientation on S, thus our marking φ is orientation preserving when the pair of homology classes $\varphi_*[\alpha], \varphi_*[\beta]$ agrees with this induced orientation, this is exactly when $A^{-1}B \in \mathbb{H}$.[1]

Given a point in $(S, \varphi) \in \mathbf{T_1}$, now identified with $(\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}), \varphi')$, there is a canonical abelian differential given by dz. Using our fixed covering $\mathbb{R}^2 \to \mathbb{T}^2$,

[1]The identification of $\mathbf{T_1}$ with lattices follows roughly [FM15]

we find that $\varphi'^*(dz) = dx + \tau dy$ our desired abelian differential. Moreover, this gives a natural identification $\mathbf{T_1} \longleftrightarrow \mathbb{H}$ by taking $\mathbb{H}$ coordinate τ.

Definition 2. *A quadratic differential on a Riemann surface S is a section of the symmetric square of the holomorphic cotangent bundle $T^{1,0} \otimes T^{1,0}$. For a quadratic differential q, its norm is defined as $||q|| = \int_S |q|$, and the space of unit quadratic differentials on S is $\mathcal{Q}^1(S)$ is the set of quadratic differentials with norm 1. We define $\mathcal{Q}^1 = \{(S, \varphi, q) \mid (S, \varphi) \in \mathbf{T_1} \text{ and } q \in \mathcal{Q}^1(S)\}$.*

Consider a point $(S, \varphi) \in \mathbf{T_1}$, the coordinate charts on S are given by translation, thus for any two coordinates we have some constant so that $z_\alpha = z_\beta + c$. A holomorphic quadratic differential has a local expression $\phi_\alpha(z_\alpha)(dz_\alpha)^2$, and the transition laws for $T^{1,0} \otimes T^{1,0}$ are $\phi_\beta(z_\beta) = \phi_\alpha(z_\alpha) \left(\frac{dz_\alpha}{dz_\beta}\right)^2$, thus $\phi_\alpha(z_\alpha) = \phi_\beta(z_\beta)$ and since S is compact the maximum principle implies $\phi \in \mathbb{C}$ is a constant. Now for $\phi(dz)^2$ to be a unit quadratic differential, we require $|\phi| = \Im(\tau)^{-1}$ so that

$$1 = \int_S |q| = |\phi| \int_S |dz|^2 = |\phi| \,\text{Area}(S) = |\phi| \,\Im(\tau)$$

Identifying (S, φ) with its torus representative as above, and using our fixed covering for $\mathbb{T}^2$, we can associate each quadratic differential q to $\varphi^*(q) = \frac{e^{i\theta}}{\Im(\tau)}(dx + \tau dy)^2$, note that this gives a natural identification of $\mathcal{Q}^1 \longleftrightarrow T^1\mathbb{H}$ since each quadratic differential is uniquely determined by (θ, τ). Since quadratic differentials on the torus are nonvanishing, we may take a locally well defined square root function, which allows us to associate 1-forms to q by considering $\Re(\sqrt{q}), \Im(\sqrt{q})$, where $\sqrt{q} = \sqrt{\phi} dz$. These naturally give rise to a foliation of the torus by considering their level sets, as well as a isotopy invariant measure for transverse curves by integrating their variation.

Definition 3. *A measured foliation of the torus is a foliation $\mathcal{F}$ of $\mathbb{T}^2$ into curves, equipped with a transverse measure μ on curves transverse to $\mathcal{F}$ satisfying μ is a locally finite borel measure on transverse curves and if I is an isotopy of curves such that $I([a, b], t)$ is transverse to $\mathcal{F}$ for all t, then $\mu(I([a, b], -))$ is constant. We identify two measured foliations if they are related by a homotopy of the torus.*

Definition 4. *We can define the space of projectivized measured foliations $\mathbb{P}\mathscr{MF}$ as $\mathscr{MF}/\mathbb{R}_{>0}$ where $\mathbb{R}_{>0}$ acts via $\lambda \cdot (\mathcal{F}, \mu) = (\mathcal{F}, \lambda\mu)$*

Measured foliations on $\mathbb{T}^2$ may be associated with $(H^1(\mathbb{T}^2; \mathbb{R}) \setminus \{0\} /\pm) \cong \mathbb{RP}^1 \times \mathbb{R}_{>0}$. To see this, we take a basis η, ξ for $H^1(\mathbb{T}^2; \mathbb{R})$ satsfying

$$\int_\alpha \eta = 1 \quad \int_\beta \eta = 0 \quad \int_\alpha \xi = 0 \quad \int_\beta \xi = 1$$

Then we associate $\mathcal{F}$ to the unique (up to sign) cohomology class ω satisfying

$$i(\mathcal{F}, n\alpha + m\beta) = \left| \int_{n\alpha + m\beta} \omega \right|$$

writing ω in our basis, we associate $\mathcal{F}$ to $|b\eta + a\xi|$ for some $a, b \in \mathbb{R}$, similarly any nonzero cohomology class considered up to sign recovers a measured foliation $(\mathcal{F}, \mu)$ since by definition the leaves of our foliation are given by $\ker|\omega|$, with transverse measure $|\omega|$. Using our fixed covering $\mathbb{R}^2 \to \mathbb{T}^2$, we identify $(\eta, \xi) \longleftrightarrow (dx, dy)$, so that compatible with our notion of pullback for abelian and quadratic differentials each measured foliation can be written as $bdx + ady$ for some $a, b \in \mathbb{R}$. We also fix an isomorphism of $\mathscr{MF} = (H^1(\mathbb{T}^2; \mathbb{R}) \setminus \{0\} / \pm)$ with $(\mathbb{RP}^1 \times \mathbb{R}_{>0})$, this is done by taking $(a, b) \mapsto b\eta + a\xi$, which we identify with $bdx + ady$, we will see later that this convention gives $\mathrm{SL}(2, \mathbb{R})$ equivariance and thus is the correct convention to agree with Ledrappier's paper. Under this identification for $X \in \mathscr{MF}$ we find $[X] \in \mathbb{P}\mathscr{MF}$ is the projection to the $\mathbb{RP}^1$ coordinate, and we can write the projection to the radial coordinate either as $||X||$ or r depending on which is more convenient. We will later see $[X]$ defines a point in the boundary of Teichmüller space consistent with $[a/b] \in \mathbb{RP}^1$.

Definition 5. *The mapping class group of the torus is the set of orientation preserving diffeomorphisms $\mathbb{T}^2 \to \mathbb{T}^2$ with the relation $\gamma \sim \gamma'$ when they are homotopic. We denote the mapping class group of the torus as Γ.*

The mapping class group acts naturally on $\mathbf{T_1}, \mathscr{MF}$, and $\mathcal{Q}^1$, these actions are respectively described as $\gamma : (S, \varphi) \mapsto (S, \varphi\gamma^{-1})$ and the latter as $(\mathcal{F}, \mu) \mapsto (\gamma(\mathcal{F}), \gamma_*\mu)$, where for a transverse curve c, $\gamma_*\mu(c) = \mu(\gamma^{-1}(c))$ is the pushforward measure, and finally for a quadratic differential $q = \phi(dz)^2$ over a surface $S \in \mathbf{T_1}$ we simply get $((S, \varphi), q) \mapsto ((S, \varphi\gamma^{-1}), q)$. In order to work with measured foliations and relate them to unit quadratic differentials, we would like to use the cohomological description, thus we would like to compute explicitly the Γ action on $H^1(\mathbb{T}^2; \mathbb{R})$.

Since mapping classes on the torus are defined up to homotopy, a mapping class may be written as a map on $H_1(\mathbb{T}^2; \mathbb{Z})$, since these are invertible and orientation preserving, the resulting matrix lies in $\mathrm{SL}(2, \mathbb{Z})$. Mapping class groups are in general generated by Dehn twists see [Mar25]. A Dehn twist about a curve c is defined as taking a tubular neighborhood D with a (orientation preserving) diffeomorphism to $r : D \to S^1 \times [-1, 1]$, and a smooth bump function $\chi : [-1, 1] \mapsto [0, 1]$ which is zero near -1 and one near 1, then defining $\gamma : \mathbb{T}^2 \to \mathbb{T}^2$ via $x \mapsto r^{-1}(e^{2\pi i \chi} r(x))$ on D and identity elsewhere. From which we may identify the Dehn twist about α as the matrix $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ and the dehn twist about β as $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, since these two matrices generate $\mathrm{SL}(2, \mathbb{Z})$ we find that $\Gamma \cong \mathrm{SL}(2, \mathbb{Z})$. This is a clean way to formally do the identification, but the identification with $\mathrm{SL}(2, \mathbb{Z})$ giving the action on $\mathcal{Q}^1$ is given in definition 6.

Now, given a matrix $g \in \mathrm{SL}(2, \mathbb{Z})$, associated to γ_*, we find that g acts on $H^1(\mathbb{T}^2; \mathbb{R})$ via left multiplication by g^{T}, which is seen from the following duality relation, which is well defined since integration is defined up to homology classes by Cauchy's theorem.

$$\int_c \gamma^*\omega = \int_{S^1} c^*\gamma^*\omega = \int_{S^1} (\gamma c)^*\omega = \int_{\gamma_* c} \omega$$

Since we have associated both $\mathbf{T_1}$ and $\mathscr{MF}$ to cohomology classes, this definition suffices to describe both mapping class group actions since the $\mathbf{T_1}$ action takes $\varphi^* dz \mapsto \varphi^*\gamma^* dz$. Using this computation and the earlier definitions for $\mathrm{SL}(2,\mathbb{Z})$ actions we establish the following convention for the left action of $\mathrm{SL}(2,\mathbb{Z})$ and $\mathrm{SL}(2,\mathbb{R})$ on $\mathcal{Q}^1$.

Definition 6. *Let* $\begin{pmatrix} r & s \\ t & u \end{pmatrix} = g \in \mathrm{SL}(2,\mathbb{Z})$*, then there is some* $\gamma \in \Gamma$ *such that* γ_* *acts on* $H_1(\mathbb{T}^2)$ *by the matrix* $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} g \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ *with respect to the basis* α, β*, then we define the left* $\mathrm{SL}(2,\mathbb{Z})$ *action* $g \cdot q$ *by* $\gamma \cdot q$*, this gives a well defined action since we are simply applying an automorphism to the identification coming from homology. Computing explicitly, take* $((S,\varphi), q)$ *and assume that* $\varphi^*\sqrt{q} = \frac{e^{\frac{i\theta}{2}}}{\sqrt{\Im(\tau)}}(dx + \tau dy)$*, so that*

$$\begin{aligned}
\sqrt{g \cdot q} = (\varphi\gamma^{-1})^* \sqrt{q} &= (\gamma^{-1})^* \frac{e^{\frac{i\theta}{2}}}{\sqrt{\Im(\tau)}}(dx + \tau dy) \\
&= \frac{e^{\frac{i\theta}{2}}}{\sqrt{\Im(\tau)}} \left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} g^{-1} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right)^* (dx + \tau dy) \\
&= \frac{e^{\frac{i\theta}{2}}}{\sqrt{\Im(\tau)}} \begin{pmatrix} u & s \\ t & r \end{pmatrix}^* (dx + \tau dy) \\
&= \frac{e^{\frac{i\theta}{2}}}{\sqrt{\Im(\tau)}} ((t\tau + u)dx + (r\tau + s)dy)
\end{aligned}$$

Which we identify with

$$\frac{e^{\frac{i\theta}{2}}(t\tau + u)}{\sqrt{\Im(\tau)}} \left(dx + \frac{(r\tau + s)}{(t\tau + u)} dy \right)$$

Thus we define the left action

$$\begin{pmatrix} r & s \\ t & u \end{pmatrix} \cdot q = \frac{e^{i\theta}(t\tau + u)^2}{\Im(\tau)} \left(dx + \frac{(r\tau + s)}{(t\tau + u)} dy \right)^2$$

We may extend this definition of the left action to all matrices in $\mathrm{SL}(2,\mathbb{R})$ *by using the same entry-wise definition.*

3 Teichmuller Metric and Dilations

Definition 7. *Given* $q \in \mathcal{Q}^1$ *lying above* $(S,\varphi) \in \mathbf{T_1}$ *we say that local coordinates* (x, y) *are natural coordinates when the horizontal and vertical foliations are given by* $|dy|$ *and* $|dx|$ *respectively.*

Existence of natural coordinates on the torus is obvious, since given the lattice presentation of a torus, and $q = \phi(dz)^2$ we simply need to multiply the lattice coordinates by $\phi^{\frac{1}{2}}$ since our differential has no zeroes we can take such a well defined branch of the square root function. This allows us to define Teichmüller mappings, which are mappings that stretch a surface along a pair of foliations.

Definition 8. *A map $h : S \to S'$ of Riemann surfaces is called Teichmüller when there are quadratic differentials q, q' on S, S' respectively such that for some $K \in \mathbb{R}_{>0}$ there are natural coordinates (x, y) near any point p and (x', y') near p' such that*

$$h(x + iy) = \sqrt{K}x' + \frac{i}{\sqrt{K}}y'$$

Such a map is said to have horizontal stretch factor $\sqrt{K}$ and dilation $\max\left\{K, K^{-1}\right\}$.

By Teichmüller's theorem [FM15] every homotopy class of maps has a unique Teichmüller mapping, given two $(S, \varphi), (S', \varphi') \in \mathbf{T_1}$ the homotopy class is always taken to be the change of markings $\varphi'\varphi^{-1}$. This allows us to define a distance function on Teichmüller space, in terms of the dilation of a Teichmüller mapping between the marked surfaces.

Definition 9. *The Teichmüller distance between two surfaces is defined as*

$$d_{\text{Teich}}(S, S') = \frac{1}{2} \log K$$

Where K is the dilation of a Teichmüller map in the homotopy class of the change of markings.

The Teichmüller distance gives a complete metric on $\mathbf{T_1}$ and the geodesics are in bijection with $\mathcal{Q}^1$, this is defined as taking $q \in \mathcal{Q}^1$ with natural coordinates (x, y), then the Teichmuller geodesic $c(t)$ is given by taking the unit quadratic differential with natural coordinates $(t^{-1}x, ty)$ [FM15].

Our goal is to understand the distribution of mapping class group orbits under the horocycle flow, which is related to the amount of shear a mapping class group element applies. Thus we will measure the size of a mapping class group element by the amount of shearing it applies to a surface.

Definition 10. *Let (S, φ) be the point in $\mathbf{T_1}$ with coordinate pulling back to $dx + idy$, then for $\gamma \in \Gamma$ we define K_γ to be the dilation of the Teichmüller map $h : S \to S$ in the homotopy class of the change of marking between (S, φ), and $\gamma \cdot (S, \varphi)$ i.e. the homotopy class of $\varphi\gamma^{-1}\varphi^{-1}$. For $g \in \mathrm{SL}(2, \mathbb{Z})$ we define K_g as K_γ for γ associated to g.*

Finally we define a section from $\mathscr{MF}$ to $\mathcal{Q}^1$, using our fixed covering of $\mathbb{T}^2$ as a basepoint, then for $\mathcal{F} \in \mathscr{MF}$ we will find that there is a unique $\mathcal{F}'$ with $[\mathcal{F}'] = [\mathcal{F}] \in \mathbb{P}\mathscr{MF}$ so that $\mathcal{F}'$ is the horizontal foliation for some $q \in \mathcal{Q}^1$ lying above dz. Thus we can recover a point in $\mathcal{Q}^1$ by travelling along the Teichmüller line associated to q until we get a quadratic differential with horizontal foliation $\mathcal{F}$. As discussed previously each quadratic differential is uniquely associated to one of the form $\frac{e^{i\theta}}{\Im\tau}(dx+\tau dy)^2$, thus fixing $\tau = i$ gives

$$\begin{aligned}\Im(\sqrt{q}) &= \sin(\theta/2)dx + \cos(\theta/2)dy \\ \Re(\sqrt{q}) &= \cos(\theta/2)dx - \sin(\theta/2)dy\end{aligned}$$

Thus for the measured foliation associated to $X = (a,b)$, we consider $q \in \mathcal{Q}^1$ with $\mathcal{F}$ with horizontal foliation $||X||^{-1}(bdx + ady)$ and vertical foliation $||X||^{-1}(adx - bdy)$, then shearing gives us a quadratic differential q with horizontal and vertical foliations given respectively by $bdx + ady$ and $||X||^{-2}(adx - bdy)$. So that the resulting $q \in \mathcal{Q}^1$ satisfies

$$\sqrt{q} = (a||X||^{-2} + ib)\left(dx + \frac{ai - ||X||^{-2}b}{bi + ||X||^{-2}a}dy\right)$$

Definition 11. *Written explicitly we get*

$$\begin{aligned}&\Psi : \mathbb{RP}^1 \times \mathbb{R}_{>0} \to \mathcal{Q}^1 \\ &(a,b) \mapsto (a||X||^{-2} + ib)^2\left(dx + \frac{ai - ||X||^{-2}b}{bi + ||X||^{-2}a}dy\right)^2\end{aligned}$$

The last preliminary notion to understand is the Teichmüller horocycle flow, however it will be far more illustrative to understand this as the the unipotent component of the $\mathrm{SL}(2,\mathbb{R})$ action on $\mathcal{Q}^1$. Let $g \in \mathrm{SL}(2,\mathbb{R})$, then the action of g on $\mathcal{Q}^1$ is simply the matrix action on the natural coordinates, using the definition as given in [For06]. Let (x', y') be natural coordinates for quadratic differential q, then in order to keep conventions consistent, we take the action on natural coordinates as $g\begin{pmatrix} y' \\ x' \end{pmatrix}$, thus the action on cohomology classes is given by the pullback, so that the action is given explicitly as

$$\begin{pmatrix} \Im\sqrt{q \cdot g} \\ \Re\sqrt{q \cdot g} \end{pmatrix} = g^{\mathrm{T}} \begin{pmatrix} \Im\sqrt{q} \\ \Re\sqrt{q} \end{pmatrix}$$

The Teichmüller unipotent flow is given by the action of the unipotent subgroup $\left\{ h_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} \mid s \in \mathbb{R} \right\}$, moreover this realizes the shearing used to define the map Ψ as the action of the diagonal subgroup.

The final piece in being able to use the geometry of $\mathbb{H}$ to describe the Teichmüller dynamics is that $(\mathbf{T_1}, 2d_{\text{Teich}})$ is isometric to $(\mathbb{H}, d_{\mathbb{H}})$, moreover this isometry respects the $\mathrm{SL}(2,\mathbb{R})$ action, in particular the Teichmüller unipotent flow is taken to the horocycle flow [FM15].

Now we give a complete coordinate description of $\mathcal{Q}^1$ of the form $\mathscr{MF} \times \mathbb{R}$, decomposing into shearing about a measured foliation and the unipotent flow. This is defined as taking $(X, s) \mapsto \Psi(X) \cdot h_s$ where $\Psi(X) \cdot h_s$ is the quadratic differential given horizontal and vertical foliations $\begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix} \begin{pmatrix} \Im\sqrt{q} \\ \Re\sqrt{q} \end{pmatrix}$.

To see this map gives a bijection write $X = g \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, for $g = R(\theta)D(||X||, ||X||^{-1})$ the composition of a diagonal and rotation matrix in $\mathrm{SL}(2,\mathbb{R})$. It follows that

$$\Psi(X) \cdot h_s = \Psi\left(g \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) \cdot h_s = (dx + idy)^2 \cdot gh_s$$

Now under the identification with $T^1\mathbb{H}$ this gives gh_s acting by mobius transform on $(i, (0, 1))$, every $\mathrm{SL}(2,\mathbb{R})$ matrix has a unique decomposition of the form $R(\theta)D(\lambda, \lambda^{-1})h_s$, our section Ψ is only defined up to sign so gives a bijection to $\mathrm{P\,SL}(2,\mathbb{R}) = \mathrm{SL}(2,\mathbb{R})/\pm 1$ and since the $\mathrm{P\,SL}(2,\mathbb{R})$ action is free and faithful we get a bijection. We may write the map and its inverse explicitly in coordinates, by applying the Teichmüller unipotent flow to the horizontal and vertical foliations of $\Psi(X)$, which gives horizontal and vertical foliations $bdx + ady$ and $||X||^{-2}(adx - bdy) + s(bdx + ady)$. This corresponds to the quadratic differential q with

$$\sqrt{q} = (bs + a||X||^{-2} + ib)\left(dx + \frac{as - b||X||^{-2} + ia}{bs + a||X||^{-2} + ib} dy\right)$$

Using this closed form, given a quadratic differential of the form $q = \frac{e^{i\theta}}{\Im(\tau)}(dx + \tau dy)^2$, the preimage under our coordinate map is

$$\begin{aligned}
\Im(\tau) &= \frac{1}{b^2 + (a||X||^{-2} + bs)^2} \\
\Re(\tau) &= \Im(\tau)(abs^2 - ab||X||^{-4} + ab - b^2 s||X||^{-2} + a^2 s||X||^{-2}) \\
\theta &= 2\arg(bs + a||X||^{-2} + ib)
\end{aligned}$$

Now we have left and right actions of $\mathrm{SL}(2,\mathbb{R})$ on $\mathcal{Q}^1$. We may see from our earlier definition that the left action is consistent with the action on $T^1\mathbb{H}$, which lies above the Möbius transform, we also note here that the left and right action agree at $\Psi(1, 0)$. The left and right action moreover commute, since our $\Re(\sqrt{q})$ and $\Im(\sqrt{q})$ are cohomology classes, and our right action gives a linear combination of them, taking the pullback before or after taking the linear combination does not affect the result.

Definition 12. *Let $g \in \mathrm{SL}(2,\mathbb{Z})$ and $X \in \mathbb{RP}^1 \times \mathbb{R}_{>0}$. Then we can define $s(g, X)$ by the equation*

$$\Psi(gX) = g \cdot \Psi(X) \cdot h_{s(g,X)}$$

To prove such an s exists, we use the KAN decomposition on $\mathrm{SL}(2,\mathbb{R})$, which gives a unique decomposition of any $\mathrm{SL}(2,\mathbb{R})$ matrix in the form $R(\theta)D(t, t^{-1})h_s$, notice by our definition of the right action if $g' \in R(\theta)D(t, t^{-1})$, then

$$\Psi \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot g' = \Psi\left(g' h_s \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right)$$

. *Now define* $g_X \in R(\theta)D(t,t^{-1})$ *as the matrix satisfying* $g_X \begin{pmatrix}1\\0\end{pmatrix} = X$. *Then we get for* KAN *decomposition* $gg_X = rdh_t$

$$\Psi(gX) = \Psi\left(gg_X\begin{pmatrix}1\\0\end{pmatrix}\right) = \Psi\begin{pmatrix}1\\0\end{pmatrix} rd = \Psi\begin{pmatrix}1\\0\end{pmatrix} gg_X h_{-t} = gg_X\Psi\begin{pmatrix}1\\0\end{pmatrix} h_{-t}$$
$$= g\Psi(X)h_{-t}$$

and thus $-t = s(g,X)$.

4 Measured Laminations

5 Projective Measured Foliations as $\partial\mathbf{T_1}$

Earlier we discussed how the $\mathbb{RP}^1$ coordinate of the measured foliation specifies a projective measured foliation, with data equivalent to the slope of a line on the torus. We can give a more precise identification with $\partial\mathbf{T_1}$ in a way that is compatible with the notion of Teichmüller geodesics. For the purposes of understanding Ledrappier's paper, we can simply think of the topology on $\mathbf{T_1}$ as being defined by the Teichmüller metric. However, to understand this phenomena we use that the topology on the Teichmüller space given equivelantly by embedding into the space of geodesic currents $\mathcal{C}$ with the weak-$*$ topology, as this is somewhat tangential I will not provide the details, and the reader is referred to [Mar25] for the proof in the negative Euler characteristic case. The embedding on the torus is given by taking (S,φ) to the functional on closed geodesics $c \mapsto \ell_S(\varphi(c))$ (here we are taking a geodesic representative of $\varphi(c)$ which is unique up to translation) the length of the curve in S. We similarly have that measured foliations have a natural mapping to the space of currents, by taking $\mathcal{F}$ given by $|\omega|$ to $c \mapsto \int_c |\omega|$. Now we can further consider the projection $\mathcal{C} \to \mathbb{P}\mathcal{C}$ which will allow us to identify $\partial\mathbf{T_1} \subset \mathcal{C}$ with $\mathbb{P}\mathscr{M}\mathscr{F}$. To do so consider a sequence $S_n \in \mathbf{T_1}$ which leaves every compact set, using the isometry with $\mathbb{H}$ each S_n corresponds to some $\tau_n \in \mathbb{H}$, and we see that under the embedding to currents S_n maps to the current $e\alpha + f\beta \mapsto (\Im(\tau))^{-\frac{1}{2}}|e + f\tau|$. One way for τ_n to leave every compact set is to have after passing to a subsequence $|\tau_n| \to \infty$, in which case we get a sequence of representatives for $\{S_n\} \to \mathbb{P}\mathcal{C}$ given by $e\alpha + f\beta \mapsto \frac{\sqrt{\Im(\tau_n)}}{|\tau_n|}\left|\frac{e+f\tau_n}{\sqrt{\Im(\tau)}}\right|$, which clearly converges to $e\alpha + f\beta \mapsto |f|$, the projective measured foliation given by $[|dy|]$. Now if $|\tau_n| \not\to \infty$ for every subsequence, then we must have $\Im(\tau_n) \to 0$, moreover after passing to a subsequence $\Re(\tau_n) \to R$ for some $R \in \mathbb{R}$ by Bolzano Weierstrass in which case we get representatives for $\{S_n\} \to \mathbb{P}\mathcal{C}$ given by $e\alpha + f\beta \mapsto \sqrt{\Im(\tau)}\left|\frac{e+f\tau_n}{\sqrt{\Im(\tau_n)}}\right|$, which converges to $e\alpha + f\beta \mapsto |e + fR|$, which corresponds to the projective measured foliation given by $[|dx + Rdy|]$. We can see this is compatible with Teichmüller geodesics since if we take a Teichmüller geodesic with horizontal foliation given by $|\omega|$ and vertical foliaton $|\nu|$, the corresponding length function becomes $c \mapsto \sqrt{\left(\int_\gamma |\omega|\right)^2 + \left(\int_\gamma |\nu|\right)^2}$, along the Teichmüller geodesic we take $(t^{-1}|\nu|, t|\omega|)$ as $t \to \infty$, thus the projective class converges to $[|\omega|]$.

6 Statement of Theorem

Let $f \in C_c(\mathbb{RP}^1 \times \mathbb{R}_{>0})$, and suppose that $[X] \in \mathbb{RP}^1$ has irrational slope, then

$$\frac{24}{\pi^2||X||}\int_{\mathbb{R}P^1\times\mathbb{R}_{>0}} f dr d\theta = \lim_{T\to\infty}\frac{1}{T}\sum_{\substack{\gamma\in\Gamma\\ \sqrt{K_\gamma}\leq T}} f(\gamma X)$$

7 Proof of Theorem

Let $X \in \mathbb{RP}^1 \times \mathbb{R}_{>0}$ be a vector with irrational slope and let $f \in C_c(\mathbb{RP}^1 \times \mathbb{R}_{>0})$, then we can define a lift $\tilde{f}$ of f to $\mathcal{Q}^1$, by distributing f along the Teichmüller unipotent flow.

Definition 13. *Let $\epsilon > 0$, then let χ_ϵ be a continuous function on $\mathbb{R}$ supported on $(-\epsilon, \epsilon)$ with $\int_{\mathbb{R}} \chi_\epsilon = 1$. Then for $q \in \mathcal{Q}^1$, bijectivity of the coordinate construction gives a unique s, such that $q \cdot h_s$ lies in $\mathrm{Im}(\Psi)$ use this to define*

$$\tilde{f}(q) = f(\Psi^{-1}(q \cdot h_s))\chi_\epsilon(s)$$

We may further define π to be the quotient map $\mathcal{Q}^1 \to \Gamma \setminus \mathcal{Q}^1$ and

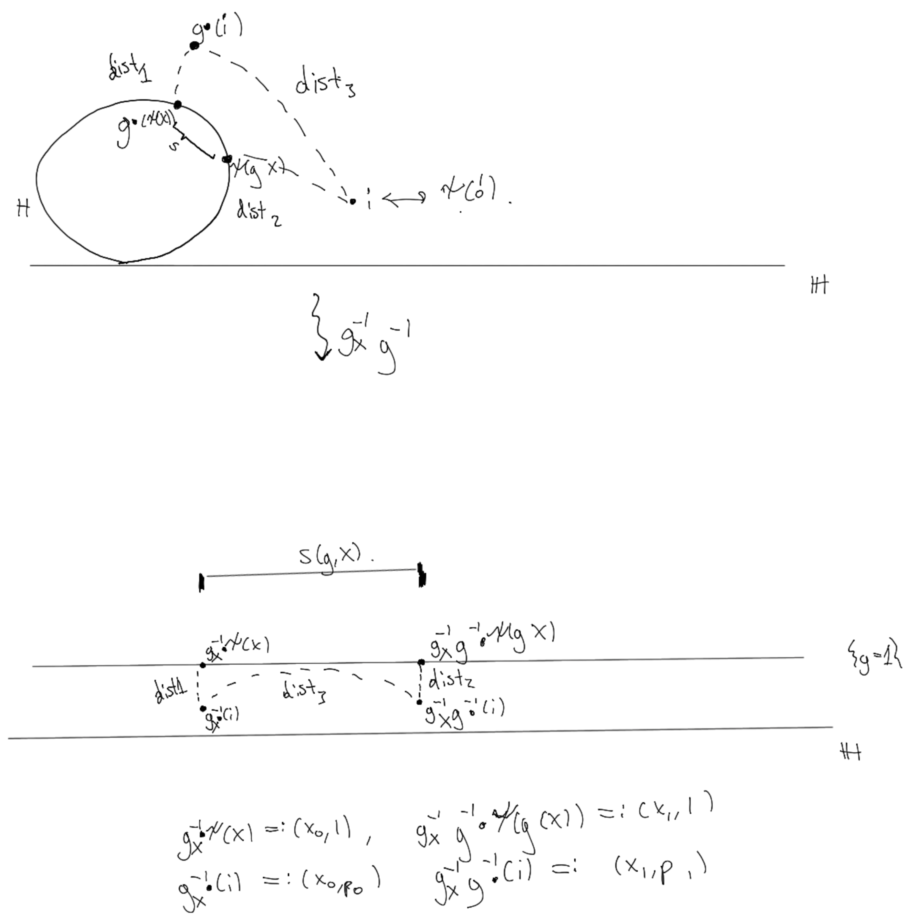
$$\begin{aligned}\overline{f} &: \mathrm{SL}(2,\mathbb{Z}) \setminus \mathcal{Q}^1 \to \mathbb{R}\\ \pi(q) &\mapsto \sum_{g\in \mathrm{SL}(2,\mathbb{Z})} \tilde{f}(g\cdot q)\end{aligned}$$

By construction, integrating $\bar{f}$ along the unipotent flow gives the sum over $\{g \in \mathrm{SL}(2,\mathbb{Z}) \mid g \cdot X \in \mathrm{supp}(f)\}$ with $s(g, X)$ lying within the integral bounds up to some small error, by relating this to the average of $f(gX)$ with K_g small, we get a new formula for the sum from the equidistribution of the unipotent flow. We can use the hyperbolic geometry to relate these two quantities as follows

Lemma 1.

$$s(g,X)^2 = \frac{K_g + K_g^{-1}}{||X||^2||gX||^2} - \frac{1}{||X||^4} - \frac{1}{||gX||^4}$$

Proof. We derive the following entirely in terms of the hyperboic geometry, firstly we are only concerned with the Teichmüller coordinate, so in this proof a quadratic differential will instead specify its projection to $\mathbf{T_1}$, moreover we use the identification with hyperbolic space, recall that under this identification we are sending $\Psi(1,0)$ to i, then all other mappings can be derived from $g\cdot\Psi(1,0)$ being identified to $g\cdot i$ where the action on $\mathbb{H}$ is Möbius transform. We abuse notation further, by writing a quadratic differential to mean its associated point in $\mathbb{H}$. Now let $g_X \in R(\theta)D(t,t^{-1})$ be the matrix satisfying $g_X\begin{pmatrix}1\\0\end{pmatrix} = X$, and let $H = gg_X\cdot\{y=1\}$, which by the equivariance lemma is the horocycle containing both $\Psi(gX)$ and $g\cdot\Psi(X)$. We get the points on the following picture, which we apply $g_X^{-1}g^{-1}$ to in order to send H to $\{y=1\}$ by isometry.



We can compute dist_1 and dist_2 using the definition d_{Teich}

$$\mathrm{dist}_1 = d_{\mathbb{H}}(g\cdot i, g\cdot\Psi(X)) = d_{\mathbb{H}}(i,\Psi(X)) = 2d_{\mathrm{Teich}}(\Psi(1,0),\Psi(X)) = \pm\log||X||^2$$

and similarly, $\mathrm{dist}_2 = \pm\log||\gamma X||^2$. Now we can use the diagram and the upper half plane distance formula $\cosh d_{\mathbb{H}}(z,w) = 1 + \frac{|z-w|^2}{2\Im(z)\Im(w)}$ applied to $z=(x_0,p_0)$

and $w = (x_1, p_1)$, from the diagram we deduce the following

$$s(g,X)^2 = \mathrm{dist}_3 = (x_1 - x_0)^2 = 2p_0p_1 \cosh d_{\mathbb{H}}(i, g\cdot i) - p_0^2 - p_1^2$$

Moreover, we know that $d_{\mathbb{H}}((x_0,1),(x_0,p_0)) = \mathrm{dist}_1$, with the distance being recorded by the vertical geodesic. This gives us

$$\begin{aligned} \mathrm{dist}_1 &= \int_1^{p_0} \frac{dy}{y} = |\log(p_0)| \\ p_0 &= e^{\pm\,\mathrm{dist}_1} = e^{\pm \log ||X||^2} = ||X||^{\pm 2} \end{aligned}$$

With the $\pm$ depending on wether or not p_0 lies above or below $\{y = 1\}$. Now since $(x_0, p_0) = g_X^{-1} \cdot i$ and $X = g_X \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ we see that p_0 is sent in the opposite direction of the stretching on X, which is sent up along the vertical trajectory. Thus $p_0 = ||X||^{-2}$ and by a similar computation $p_1 = ||\gamma X||^{-2}$. Substituting in these values and using that $\cosh(d_{\mathbb{H}}(i, g\cdot i)) = \frac{K_g + K_g^{-1}}{2}$ we get the desired result. □

Now since f has compact support, the $\frac{1}{||gX||^4}$ term is bounded for $gX \in \mathrm{supp}(f)$ and also note that $K_g \geq 1$, so that the $\frac{K_g^{-1}}{||X||^2||gX||^2}$ is also uniformly bounded for all g, this furnishes positive constants C and D such that

$$\frac{K_g}{||X||^2||gX||^2} - C \leq s(g,X)^2 \leq \frac{K_g}{||X||^2||gX||^2} + D$$

whence we find that

$$\begin{aligned} |s(g,X)| &\leq \frac{T}{||X||||gX||} - \sqrt{C} \\ \implies \sqrt{K_g} &\leq ||X||||gX|| \left(\sqrt{\frac{K_g}{||X||^2||gX||^2} - C} + \sqrt{C} \right) \leq T \end{aligned}$$

and if $\sqrt{K_g} \leq T$, then

$$|s(g,X)| \leq \sqrt{\frac{T^2}{||X||^2||gX||^2} + D} \leq \frac{T}{||X||||gX||} + \sqrt{D}$$

The first thing to notice from these equations is that $\overline{f}$ is well defined, this follows immediately from the fact that Γ as a mapping class group acts properly discontinuously [FM15], which having infinitely many $\gamma \in \Gamma$ with $K_\gamma \leq N$ for some N would directly contradict. There is also a way to see this simply using the norm on $\mathrm{SL}(2,\mathbb{Z})$.

$$K_g + \frac{1}{K_g} = 2\cosh 2d_{\mathrm{Teich}}(S, g\cdot S) = 2\cosh d_{\mathbb{H}}(i, g\cdot i) = ||g||_F^2$$

So that for any N there are finitely many γ with $|s(\gamma, X)| \leq N$. Either approach implies $\bar{f}$ is continuous since it is given locally by a sum of continuous functions. The second application of these inequalities is defining $L = \inf_{Y \in \mathrm{supp}(f)} ||Y||$ and $M = \sup_{Y \in \mathrm{supp}(f)} ||Y||$ we get the following explicit relation for positive valued f,

$$\int_{-\frac{T}{||X||M}+\sqrt{D}+\epsilon}^{\frac{T}{||X||M}-\sqrt{D}-\epsilon} \overline{f}(\pi\Psi(X)\cdot h_s)ds \leq \sum_{\substack{\gamma\in\Gamma \\ \sqrt{K_\gamma}\leq T}} f(\gamma X) \leq \int_{-\frac{T}{||X||L}-\sqrt{C}-\epsilon}^{\frac{T}{||X||L}+\sqrt{C}+\epsilon} \overline{f}(\pi\Psi(X)\cdot h_s)ds$$

$$\lim_{T\to\infty} \frac{1}{T} \int_{-\frac{T}{||X||M}}^{\frac{T}{||X||M}} \overline{f}(\pi\Psi(X)\cdot h_s)ds \leq \lim_{T\to\infty} \frac{1}{T} \sum_{\substack{\gamma\in\Gamma \\ \sqrt{K_\gamma}\leq T}} f(\gamma X) \leq \lim_{T\to\infty} \frac{1}{T} \int_{-\frac{T}{||X||L}}^{\frac{T}{||X||L}} \overline{f}(\pi\Psi(X)\cdot h_s)ds$$

Similar equalities holds for negative valued f. We may assume that f is strictly positive or negative, considering f^+, f^- defined as $\max\{f, 0\}, \min\{f, 0\}$ respectively. To complete the proof I will evaluate $\lim_{T\to\infty} \frac{1}{T} \int_{-\frac{T}{||X||M}}^{\frac{T}{||X||M}} \overline{f}(\pi\Psi(X)\cdot h_s)ds$, since the same argument applies to the other limit squeezing the sum and gives the same result.

Now we can see that the horocycle flow for $\Psi(X)$ is aperiodic and hence equidistributes by Dani's theorem if and only if X has irrational slope, to see this suppose for $g \in \mathrm{SL}(2,\mathbb{Z})$ and $s \in \mathbb{R}\backslash\{0\}$ we have $\Psi(X)\cdot h_s = g\cdot\Psi(X)$, then letting $g_X \in R(\theta)D(t, t^{-1})$ satisfy $g_X \begin{pmatrix} 1 \\ 0 \end{pmatrix} = X$ we find that $gg_X \cdot \Psi \begin{pmatrix} 1 \\ 0 \end{pmatrix} = g_X h_s \Psi \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and since the $\mathrm{PSL}(2,\mathbb{R})$ action is free this implies that $g_X h_s g_X^{-1} = \pm g$, then since $g_X = \begin{pmatrix} a & -b||X||^{-2} \\ b & a||X||^{-2} \end{pmatrix}$, we can compute $g_X h_s g_X^{-1} = \begin{pmatrix} 1-abs & a^2 s \\ -b^2 s & 1+abs \end{pmatrix}$, this cannot possibly be in $\mathrm{SL}(2,\mathbb{Z})$, since if it were we would require $abs \in \mathbb{Z}$ and $b^2 s \in \mathbb{Z}$ implying that $abs/b^2 s = a/b \in \mathbb{Q}$. Thus by Dani's theorem [Dan82] we have for μ the unit Haar measure on $\mathrm{SL}(2,\mathbb{Z}) \setminus \mathrm{SL}(2,\mathbb{R})$

$$\begin{aligned}
\lim_{T\to\infty} \frac{1}{T} \int_{-\frac{T}{||X||M}}^{\frac{T}{||X||M}} \overline{f}(\pi\Psi(X)\cdot h_s)ds &= \lim_{T\to\infty} \frac{1}{T||X||M} \int_{-T}^{T} \overline{f}(\pi\Psi(X)\cdot h_s)ds \\
&= \lim_{T\to\infty} \frac{1}{T||X||M} \int_{-T}^{T} \overline{f}\left(\pi g_X h_s \cdot \Psi \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) ds \\
&= \frac{2}{||X||M} \int_{\mathrm{SL}(2,\mathbb{Z})\setminus\mathrm{SL}(2,\mathbb{R})} \overline{f}\left(\pi g \cdot \Psi \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) d\mu
\end{aligned}$$

Lemma 2.

$$\int_{\mathrm{SL}(2,\mathbb{Z})\setminus\mathrm{SL}(2,\mathbb{R})} \overline{f}\left(\pi g \cdot \Psi \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) d\mu = \frac{12}{\pi^2} \int_{\mathbb{RP}^1 \times \mathbb{R}_{>0}} f(re^{i\theta}) r dr d\theta$$

Proof. We can parameterize $\mathrm{SL}(2,\mathbb{R})$ by $\left\{ \begin{pmatrix} u & w \\ v & h \end{pmatrix} \mid uh - vw = 1 \right\}$, then the integral over $\mathrm{SL}(2,\mathbb{R})$ is the integral over the chart $u \neq 0$. In this chart we have

$d\mu = \frac{6}{\pi^2}\frac{|du\, dv\, dw|}{|u|}$, in our earlier parameterization in terms of u, v, s we have $w = us - v(u^2+v^2)^{-1}$, this gives us Jacobian determinant u. Now letting D be a fundamental domain for $\mathrm{SL}(2,\mathbb{Z}) \setminus \mathrm{SL}(2,\mathbb{R})$ we get from definition of $\tilde{f}$

$$\begin{aligned}
\int_{\mathrm{SL}(2,\mathbb{Z})\setminus\mathrm{SL}(2,\mathbb{R})} \overline{f}\left(\pi g \cdot \Psi \begin{pmatrix}1\\0\end{pmatrix}\right) d\mu &= \int_D \sum_{g'\in\Gamma} \tilde{f}\left(g' g \cdot \Psi \begin{pmatrix}1\\0\end{pmatrix}\right) d\mu \\
&= \int_D \sum_{g'\in\Gamma} \tilde{f}\left(\Psi \begin{pmatrix}1\\0\end{pmatrix} \cdot g' g\right) d\mu \\
&= \int_{\mathrm{SL}(2,\mathbb{R})\cap\{u\neq 0\}} \tilde{f}\left(\Psi \begin{pmatrix}1\\0\end{pmatrix} \cdot \begin{pmatrix}u & w\\ v & h\end{pmatrix}\right) \frac{6}{\pi^2}\frac{du\, dv\, dw}{|u|} \\
&= \frac{6}{\pi^2}\int_{\mathrm{SL}(2,\mathbb{R})\cap\{u\neq 0\}} \tilde{f}\left(\Psi \begin{pmatrix}1\\0\end{pmatrix} \cdot \begin{pmatrix}u & us - v(u^2+v^2)^{-1}\\ v & vs + u(u^2+v^2)^{-1}\end{pmatrix}\right) du\, dv\, ds \\
&= \frac{12}{\pi^2}\int_{\mathbb{RP}^1\times\mathbb{R}_{>0}\times\mathbb{R}} \tilde{f}\left(\Psi\left((\theta, r)\right) \cdot h_s\right) r dr\, d\theta\, ds \\
&= \frac{12}{\pi^2}\int_{\mathbb{RP}^1\times\mathbb{R}_{>0}\times\mathbb{R}} \chi_\epsilon(-s) f(re^{i\theta}) r dr d\theta ds \\
&= \frac{12}{\pi^2}\int_{\mathbb{RP}^1\times\mathbb{R}_{>0}} f(re^{i\theta}) r dr d\theta
\end{aligned}$$

□

Combining results gives us

$$\lim_{T\to\infty}\frac{1}{T}\int_{-\frac{T}{||X||M}}^{\frac{T}{||X||M}} \overline{f}(\pi\Psi(X)\cdot h_s) ds = \frac{24}{\pi^2 ||X|| M}\int_{\mathbb{RP}^1\times\mathbb{R}_{>0}} f(re^{i\theta}) r dr d\theta$$

eliminate the $1/M$ factor, we note that f is supported on the annulus $\left\{re^{i\theta} \mid L \le r \le M\right\}$, we can take for any natural N, a continuous partition of unity of the annulus $\chi^1, \ldots, \chi^N$ with χ^j supported on $\left\{re^{i\theta} \mid L_j = L + \frac{M-L}{N}(j-2) < r < L + \frac{M-L}{N}(j+1) = M_j\right\}$, writing $f^j = \chi^j f$ we recover that

$$\sum_1^N \lim_{T\to\infty}\frac{1}{T}\int_{-\frac{T}{||X||M_j}}^{\frac{T}{||X||M_j}} \overline{f^j}(\pi\Psi(X)\cdot h_s) ds \le \lim_{T\to\infty}\frac{1}{T}\sum_{\substack{\gamma\in\Gamma\\ \sqrt{K_\gamma}\le T}}\sum_j f^j(\gamma X) = \lim_{T\to\infty}\frac{1}{T}\sum_{\substack{\gamma\in\Gamma\\ \sqrt{K_\gamma}\le T}} f(\gamma X)$$

Applying Dani's theorem and lemma 2 to the term on the left hand side we get

$$\sum_1^N \lim_{T\to\infty}\frac{1}{T}\int_{-\frac{T}{||X||M_j}}^{\frac{T}{||X||M_j}} \overline{f^j}(\pi\Psi(X)\cdot h_s) ds = \sum_1^N \frac{24}{\pi^2 ||X|| M_j}\int_{\mathbb{RP}^1\times\mathbb{R}_{>0}} f^j(re^{i\theta}) r dr d\theta$$

Then since this hold for all N, we may write the following and take $N \to \infty$

$$\left|\frac{24}{\pi^2||X||}\int_{\mathbb{R}P^1\times\mathbb{R}_{>0}} f dr d\theta - \sum_1^N \frac{24}{\pi^2||X||M_j}\int_{\mathbb{RP}^1\times\mathbb{R}_{>0}} f^j r dr d\theta\right|$$
$$=\frac{24}{\pi^2||X||}\left|\int_{\mathbb{R}P^1\times\mathbb{R}_{>0}} \sum_1^N f^j - \frac{1}{M_j} f^j r dr d\theta\right|$$
$$\leq \frac{24||f||_\infty}{\pi^2||X||}\sum_1^N \int_0^\pi \int_{L_j}^{M_j} 1 - \frac{L_j}{M_j} dr d\theta \leq \frac{12||f||_\infty}{\pi||X||L}\sum_1^N (M_j - L_j)^2$$
$$\leq \frac{24||f||_\infty}{\pi||X||L}\frac{(M-L)^2}{N}$$

The exact same proof applies to the upper bound, so that we get the desired result

$$\frac{24}{\pi^2||X||}\int_{\mathbb{R}P^1\times\mathbb{R}_{>0}} f dr d\theta = \lim_{T\to\infty}\frac{1}{T}\sum_{\substack{\gamma\in\Gamma \\ \sqrt{K_\gamma}\leq T}} f(\gamma X)$$

8 References

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